



Debugging guide

How to find out what's wrong and fix it: the doctor, the logs, ways to narrow a problem down, and a symptom-by-symptom guide.

Jarvis 1.0, October 2026

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Start here: the doctor

```
./run.sh --doctor
```

It runs every check in a few seconds, recording 1 second from the microphone and sending one tiny message to Claude. Fix the lines marked **X** first, then those marked **!**. Lines marked **.** are information.

Section	What it checks	Usual fix
System	macOS version, processor, Python 3.11 or 3.12, running inside .venv	Install python@3.12 and run ./setup.sh; always start Jarvis with ./run.sh.
Python packages	Each package imports	Run ./setup.sh again.
Settings	Files loaded, unknown setting names, the model, the workspace	Fix the name in config.local.yaml.
Mac permissions	Accessibility and Screen Recording	System Settings, Privacy & Security.
Hotkeys	Each hotkey and the hold-to-talk key	Use keys with modifiers, or a function key.
Microphone and speech	Speech model, speaking voice, microphone name and level	Microphone permission; voice.input_device.
Extras	Wake word, start at login	./setup.sh --wakeword; ./run.sh --login on.
Claude API	The key, and a real test message	Re-copy the key; add credit; check llm.model.

Logs and the debug switch

Where	What you'll find
data/logs/jarvis.log	Everything: requests, tool calls, errors with full tracebacks. The menu's Open log opens it.
./run.sh --debug ...	The same detail in the terminal as it happens, for example ./run.sh --debug --cli.
data/logs/background/*.log	Output of servers and long jobs Jarvis started in the background.
data/logs/patch-phase-N-tests.log	Test output when an update kit stopped.

```
tail -f data/logs/jarvis.log          # watch it live
grep -iE "error|warning" data/logs/jarvis.log | tail -20
./run.sh --debug --cli               # detailed logs in the terminal
```

Useful lines to look for: “Tool ... error”, “Web search unavailable”, “The floating panel couldn't start”, “The tutor pointer couldn't start”, “Hotkeys active (listen-only...)”, “Couldn't start the hotkey listener”.

Narrowing a problem down

Jarvis is a chain: microphone, speech recognition, Claude, tools, speech, screen. Test one link at a time:

To test	Run	Needs a key
Microphone and speech recognition	<code>./run.sh --listen</code>	No
Claude and the tools, without voice	<code>./run.sh --cli</code>	Yes
A single question	<code>./run.sh --say "what's 2+2"</code>	Yes
Voice end to end, without the menu bar	<code>./run.sh --cli --voice</code>	Yes
Menu bar, hotkeys and panel	<code>./run.sh</code>	Hotkeys and listening work without one
Tutor mode, without hotkeys	<code>/tutor in ./run.sh --cli</code>	Yes

- Use a cheaper model while experimenting: `./run.sh --model claude-haiku-4-5-20251001 --cli`.
- Start a new conversation (`/new` or say “new conversation”) when Jarvis seems confused by earlier messages.
- List microphones with `.venv/bin/python -m sounddevice`, and voices with `say -v '?'`.
- See what Jarvis remembers with `/memory`; memory is stored in `data/memory.json`.
- Run the tests (`.venv/bin/python -m unittest discover -s tests`) to rule out a damaged install.
- Get files back: copies made by update kits are in `patches/backups/`; files Jarvis changed outside the workspace are in `data/backups/`.

Symptoms, causes and fixes

Starting up

Symptom	Likely cause	Fix
“Jarvis needs Python 3.11 or 3.12”	No suitable Python	<code>brew install python@3.12</code> , then <code>./setup.sh</code> .
“Run <code>./setup.sh</code> first.”	No <code>.venv</code>	<code>./setup.sh</code> .
“Couldn't load the macOS app (...)”	PyObjC missing or damaged	<code>./setup.sh</code> , then <code>./run.sh --doctor</code> .
No menu-bar icon	The app stopped, or the Terminal window was closed	Read the Terminal window and the log; start <code>./run.sh</code> again and keep the window open.
“Warning: Unknown setting ...”	A typo in <code>config.local.yaml</code>	Fix the setting's name; the doctor lists them.
SSL or certificate errors	The python.org installer	Run “Install Certificates.command” from Applications, Python 3.12.

Hotkeys and the menu bar

Symptom	Likely cause	Fix
<code>^⌘C</code> does nothing	Accessibility is off for Terminal	Turn it on; Jarvis retries every 3 s for about 5 minutes, otherwise restart it.
Works in some apps only	Secure Keyboard Entry, or a password field has focus	Turn off Terminal, Secure Keyboard Entry; click out of the password field.
The letter also gets typed	Listen-only mode (only Input Monitoring allowed)	Allow Accessibility for Terminal.
A hotkey warning at start	An invalid hotkey in your settings	Use modifiers (<code>ctrl+alt+...</code>) or a function key.
Holding Option doesn't start listening	Hold-to-talk off, another key pressed, or held under 0.3 s	Check <code>hotkeys.push_to_talk</code> ; hold a little longer.
The panel never appears	The panel couldn't start, so the menu bar took over	Search the log for “floating panel couldn't start” and send it to me.

Listening and speech

Symptom	Likely cause	Fix
“The microphone is giving pure silence”	No Microphone permission	Allow Terminal, then quit and reopen Terminal.
“Didn't hear anything”	Too quiet, wrong microphone or threshold too high	Check the level in the doctor; set <code>voice.input_device</code> ; lower <code>voice.vad_threshold</code> .
Keeps listening	A noisy room	Raise <code>voice.vad_threshold</code> , for example to 0.02.
Cuts you off	Your pauses are longer than 1.2 s	Raise <code>voice.silence_seconds</code> , for example to 1.8.
Wrong words	Names, accent or a small model	Add words to <code>voice.vocabulary</code> ; try <code>medium.en</code> ; speak closer.
Slow to transcribe	A big model on an older processor	Use <code>voice.stt_model: base.en</code> .
Speech model download fails	No internet on first use	Reconnect and run <code>./run.sh --listen</code> again.
No spoken replies	<code>voice.tts</code> is off, terminal chat without <code>--voice</code> , or the volume	Check the setting; run <code>say hello</code> in Terminal.
“Voice ... isn't installed”	<code>voice.tts_voice</code> isn't on this Mac	Pick one from <code>say -v '?'</code> or download it.

Claude and the tools

Symptom	Likely cause	Fix
“My API key was rejected”	Wrong or partial key	Re-copy it into <code>.env</code> , no quotes or spaces.
“The model ... wasn't found”	A typo in <code>llm.model</code>	Use <code>claude-sonnet-5-5</code> , <code>claude-haiku-4-5-20251001</code> or <code>claude-opus-5-5</code> .

Symptom	Likely cause	Fix
"credit balance is too low"	No credit on the account	Add credit under Billing in the Claude Console.
"rate limited" or "overloaded"	Too many requests, or busy servers	Wait a minute and try again.
"can't reach the Anthropic API"	Internet, VPN or proxy	Check the connection.
"Web search unavailable" in the log	Web search is off for your organisation	Turn it on in the Console, or carry on without it.
"I've hit my step limit"	The task needed more than 30 steps	Say "continue", or raise <code>llm.max_steps</code> .
"The user declined..."	You answered no to a risky action	That's fine: ask in a different way.
"Writing to ... is blocked for safety"	A protected folder	Choose a different location.
"macOS blocked access"	Files and Folders or Full Disk Access	Allow Terminal in Privacy & Security.
AppleScript "Not authorized ... (-1743)"	Automation permission	Privacy & Security, Automation: allow Terminal for that app.
"I need Accessibility permission to type"	Typing or reading a selection	Allow Accessibility for Terminal.
"The terminal running Jarvis is in front"	The terminal chat's typing guard	Bring the right app forward, or ask Jarvis to open it first.
"Timed out after 120s"	A long command	Ask Jarvis to run it in the background, or raise <code>safety.shell_timeout_seconds</code> .

Tutor mode and extras

Symptom	Likely cause	Fix
Jarvis only sees the wallpaper	No Screen Recording permission	Allow Terminal, then quit and reopen Terminal.
The pointer is in the wrong place	Jarvis looks at the main display only (the one with the menu bar)	Use tutor mode on the main display.
No pointer at all	The pointer couldn't start	Jarvis still explains; send me the log line.
"Hey Jarvis" is ignored	Not installed, not enabled, or the threshold	Check the doctor's Extras; <code>./setup.sh --wakeword</code> ; lower <code>wake_word.threshold</code> .
Wakes up by itself	Threshold too low	Raise <code>wake_word.threshold</code> to 0.6 or 0.7.
Start at login didn't take	Terminal may not control System Events	Allow it under Automation, or add <code>Jarvis.command</code> in System Settings, General, Login Items.

Updates and tests

- If `apply.sh` stops because tests failed, nothing was committed. Read `data/logs/patch-phase-N-tests.log` and send it to me; your previous files are in `patches/backups/`.
- If the patch doesn't fit because you edited files, `apply.sh` copies files instead and keeps your versions in `patches/backups/`. Copy your changes back by hand.
- A red badge on GitHub: open the failed run in the Actions tab; the log names the failing test.
- Run one test file with `-v` to see every test and the first failure in detail.

Asking for help

When something goes wrong, send:

- What you did and what you expected, with the exact words you said or typed.
- The output of `./run.sh --doctor` and of `./run.sh --version`.
- The end of the log: `tail -50 data/logs/jarvis.log` (remove anything private).
- Anything printed in the Terminal window.

Never share your API key. It lives only in `.env`, and Jarvis never writes it to the log.